CSA1311 - Theory of Computation

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Assignment-1

(1) Design a DFA to accept strings that start and end with same symbols over $\Sigma = \{0, 1\}$. Write the formal definition of the DFA. Also show that the string 010100 is accepted by the DFA.

sol:-

Deterministic Finite Automata (DFA):-

A DFA is represented by 5-tuple:

$$M = (Q, \Sigma, \delta, q_0, F)$$

transition function is :

$$\delta : Q \times \Sigma \rightarrow Q$$

This means that given a current state and an input symbol, the DFA produces exactly one next state.

where :

- Q → finite set of states
- Σ (sigma) → Input alphabet
- δ (delta) → Transition function
- q_0 → Initial state
- F → Final / accepting states.

Designing a DFA that accept strings start and end with same symbols.

We need to design a DFA that accepts strings whose:

- First symbol is 0 and last symbol is 0, OR
- First symbol is 1 and last symbol is 1.

Examples:

String	Result
00	- Accepted
11	- Accepted

010100 - Accepted

101 - Accepted

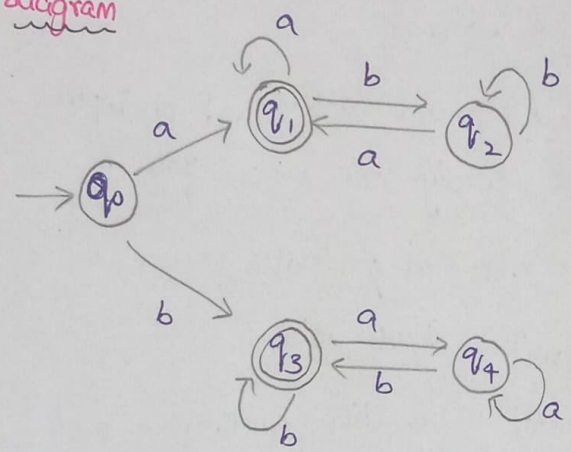
01 - Rejected.

State definitions

We require five states:

- q_0 : Initial state.
- q_1 : first symbol is 0 and current last symbol is 0.
- q_2 : first symbol is 0 and current last symbol is 1.
- q_3 : first symbol is 1 and current last symbol is 1.
- q_4 : first symbol is 1 and current last symbol is 0.

State diagram



Transition table

Current state	Input 0	Input 1
$\rightarrow q_0$	q_1	q_3
$*q_1$	q_1	q_2
q_2	q_1	q_2
$*q_3$	q_4	q_3
q_4	q_4	q_3

$\rightarrow$ is initial state, * indicates final or accepting states.

Here $Q = \{q_0, q_1, q_2, q_3, q_4\}$, $\Sigma = \{0, 1\}$

Initial state $= q_0$, Final states $\Rightarrow F = \{q_1, q_3\}$

Transition function:

$\delta(q_0, 0) = q_1$ $\quad$ $\delta(q_2, 1) = q_2$

$\delta(q_0, 1) = q_3$ $\quad$ $\delta(q_3, 0) = q_4$

$\delta(q_1, 0) = q_1$ $\quad$ $\delta(q_3, 1) = q_3$

$\delta(q_1, 1) = q_2$ $\quad$ $\delta(q_4, 0) = q_4$

$\delta(q_2, 0) = q_1$ $\quad$ $\delta(q_4, 1) = q_3$

Verification of string 010100

Input string : 010100

Step	Input	state transitions
0	-	q_0
1	0	$q_0 \to q_1$
2	1	$q_1 \to q_2$
3	0	$q_2 \to q_1$
4	1	$q_1 \to q_2$
5	0	$q_2 \to q_1$
6	0	$q_1 \to q_1$

Final state : $q_1 \in F$ $\Rightarrow$ Accepted //

Therefore, the string 010100 is accepted because it starts with 0 and end with 0.

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(2) Design an NFA to accept strings having 101 as a substring.

sol:-

Non-Deterministic Finite Automata

A string is accepted if atleast one possible path finishes in a final state after reading the entire string.

NFA transition function is :

$$\delta : Q \times \Sigma \to 2^Q$$

2^Q means power set of Q.

Designing an NFA:

We need to design an NFA that accepts all strings containing 101 anywhere in the input.

The alphabet is: $\Sigma = \{0, 1\}$

Examples:

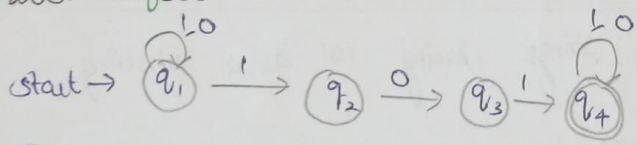
String		Result
101	–	Accepted
0101	–	Accepted
1010	–	Accepted
11101	–	Accepted
111	–	Rejected
100	–	Rejected

State definitions:

We use four states:

- q_0: Starting state; searching for the substring 101.
- q_1: First 1 has been found
- q_2: substring 10 has been found
- q_3: Complete substring 101 has been found (final state).

State Diagram:



Transition table:

State	Input 0	Input 1.
$\rightarrow q_0$	$\{q_0\}$	$\{q_0, q_1\}$
q_1	$\{q_2\}$	$\emptyset$
q_2	$\emptyset$	$\{q_3\}$
$*q_3$	$\{q_3\}$	$\{q_3\}$

Example verification: 1101

Input	Possible state progression.
start	$\{q_0\}$
1	$\{q_0, q_1\}$
1	$\{q_0, q_1\}$
0	$\{q_0, q_2\}$
1	$\{q_0, q_1, q_3\}$

since the final state contains q_2, the string is accepted.

(3) Apply the principle of mathematical induction and prove that for all positive integers n, the statement given below is true.

$$\sum_{i=1}^{n} i(i+1) = \frac{n(n+1)(n+2)}{3}$$

Sol:

Proof by mathetical induction:

We need to prove:

$$P(n): \sum_{i=1}^{n} i(i+1) = \frac{n(n+1)(n+2)}{3}$$

Mathematical induction consists of three steps:

Step 1: Base case (n=1)

Substitute n=1 in the LHS.

$$LHS = \sum_{i=1}^{1} i(i+1) = 1(1+1) = 2$$

Now substitute n=1 in the RHS

$$RHS = \frac{1(1+1)(1+2)}{3}$$

$$= \frac{1 \times 2 \times 3}{3} = 2$$

Therefore, LHS = RHS = 2

Step 2: Induction hypothesis

Assume that the statement is true for $n=k$.

That is, $$\sum_{i=1}^{k} i(i+1) = \frac{k(k+1)(k+2)}{3}$$

This is our induction hypothesis.

Step 3: Induction step

We need to prove that the statement is true for $n=k+1$

Consider the LHS: $\sum_{i=1}^{k+1} i(i+1)$

seperating the last term:

$$= \sum_{i=1}^{k} i(i+1) + (k+1)(k+2)$$

Using the induction hypothesis:

$$= \frac{k(k+1)(k+2)}{3} + (k+1)(k+2)$$

Taking common factors:

$$= (k+1)(k+2)\left[\frac{k+3}{3}\right]$$

$$= \frac{(k+1)(k+2)(k+3)}{3}$$

$$= \frac{(k+1)((k+1)+1)((k+1)+2)}{3}$$

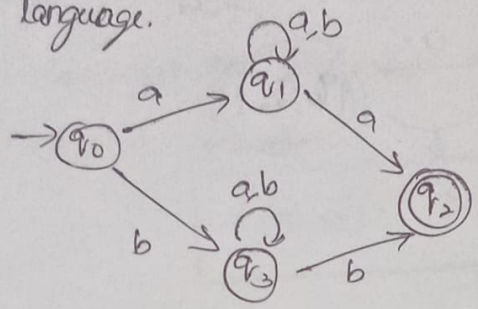
This is exactly the RHS when $n=k+1$:

Therefore, $P(k) \Rightarrow P(k+1)$

Conclusion: Since the statement is true for $n=1$, and if it is true for $n=k$, then it is also true for $n=(k+1)$.

Hence, $$\boxed{\sum_{i=1}^{n} i(i+1) = \frac{n(n+1)(n+2)}{3}}$$ is true for all positive integers n.

Consider the NFA given below which accepts the language representing the set of all strings that start and end with same symbol. over the alphabet $\Sigma = \{a,b\}$. Construct a DFA that represents the same language.



Sol:

Converting NFA to equivalent DFA:

We have given a NFA that accepts strings that start and end with same symbol over: $\Sigma = \{a,b\}$.

We need to convert this NFA into an equivalent DFA using the subset Construction Method.

Step 1: Draw the Transition table.

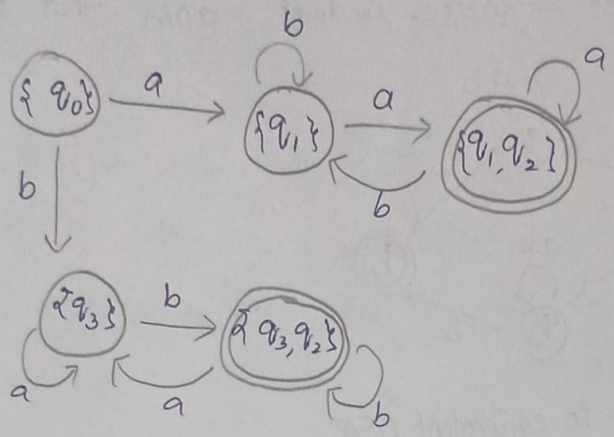
State	Input a	Input b
$\rightarrow q_0$	$\{q_1\}$	$\{q_3\}$
q_1	$\{q_1, q_2\}$	$\{q_1\}$
$*q_2$	~~ϕ~~	ϕ
q_3	$\{q_3\}$	$\{q_2, q_3\}$

Step 2: Transition table of DFA

δ	a	b
$\rightarrow q_0$	$\{q_1\}$	$\{q_3\}$
q_1	$\{q_1, q_2\}$	$\{q_1\}$
q_3	$\{q_3\}$	$\{q_1, q_3\}$
$*\{q_1, q_2\}$	$\{q_1, q_2\}$	$\{q_1\}$
$*\{q_2, q_3\}$	$\{$~~q_1, q_2~~$, q_3\}$	$\{q_2, q_3\}$

$\{q_1, q_2, q_3\}$	$\{q_1, q_2, q_3\}$	$\{q_1, q_3\}$

Step 3: DFA State Diagram

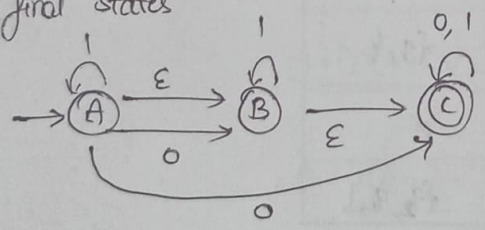


Result:- The equivalent DFA contains five states:

$q_0, q_1, q_3, q_1q_2, q_2q_3$. It accepts the same language as the given NFA.

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(5) A NFA with ε-transitions is given below. Find ε-closure for each state and construct an equivalent NFA without ε-transitions that accepts the same language. Draw its transition table and mark the initial and final states



Sol:- We need to remove all ε-transitions from the given NFA while keeping the same language.

Understand the given NFA.

The given automation has three states:

- A → Initial state
- B → Intermediate state
- C → Final state.

Step 1: Find ε-closure for each state

ε-closure (A) = {A, B, C}

ε-closure (B) = {B, C}

ε-closure (C) = {C}

Step 2: Construct an equivalent NFA without ε-transitions

Transition formula:

$$\delta'(q, a) = \varepsilon\text{-closure}\left(\bigcup_{p \in \varepsilon\text{-closure}(q)} \delta(p, a)\right)$$

1. Transitions from A $\quad E(A) = \{A, B, C\}$

for input 0:

$$\delta'(A,0) = E(\delta(E(A), 0))$$

$$\delta(E(A),0) = \delta(A,0) \cup \delta(B,0) \cup \delta(C,0)$$

$$= \{B, C\} \cup \{\emptyset\} \cup \{C\}$$

$$\delta'(A,0) = E(\{B, C\}) = \{B, C\}$$ //

for input 1:

$$\delta'(A,1) = E(\delta(E(A), 1))$$

$$= E(\delta(A,1) \cup \delta(B,1) \cup \delta(C,1))$$

$$\delta'(A,1) = E(\{A, B, C\}) = \{A, B, C\}$$ //

2. Transitions from B

$$E(B) = \{B, C\}$$

for input 0:

$$\delta'(B,0) = E(\delta(E(B), 0))$$

$$= E(\delta(B,0) \cup \delta(C,0))$$

$$\delta'(B,0) = \{C\}$$ //

for input 1:

$$\delta'(B,1) = E(\delta(E(B), 1))$$

$$\delta'(B,1) = E(\delta(B,1) \cup \delta(C,1)) = \{B, C\}$$ //

3. Transitions from C

$E(c) = \{c\}$

for input 0:

$\delta'(c,0) = \{c\}$

for input 1:

$\delta'(c,1) = \{c\}$

Step 3: Final transition table.

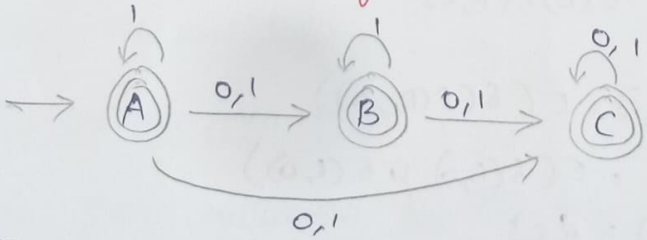
Current state	Input 0	Input 1
→ A	{B}	{A}
B	∅	{B}
*C	{C}	{C}

Step 4: Transition table of equivalent NFA (without ε-transitions)

State	0	1
→ *A	{B, C}	{A, B, C}
*B	{C}	{B, C}
*C	{C}	{C}

Since C is in the closure of A, B, and C, the final states are A, B and C (all three).

Step 5: State Diagram of ε-free NFA.



Since A is now final state, the empty string ε is accepted, which matches the original NFA (A → ε, B → ε, C reaches the final state).